

AI-Powered ATS Evaluation Engine v1

Week 3 Evaluation Report

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Project: Hiresense AI Resume Parser / ATS Engine

1. Overview

I built and tested all seven Week 3 tasks for the ATS Evaluation Engine v1. I tested everything against three real resumes (from a UF Career Connections sample packet) paired with three job descriptions - a strong match, a medium match, and a deliberately weak match - so I could check whether each component correctly told good fits apart from poor ones.

In nearly every task, my first version had a real bug that testing caught. I'm documenting those fixes below because they show the actual process, not just a final working script.

2. Task Summary

Task	Deliverable	Status
1. Improve Semantic Matching	semantic_matcher_v2.py	Complete
2. Skill Match Score	skill_matcher.py	Complete
3. Experience Matching	experience_matcher.py	Complete
4. Education Matching	education_matcher.py	Complete
5. Build ATS Score v2	ats_engine_v2.py	Complete
6. Skill Gap Analysis	skill_gap_engine.py	Complete
7. Evaluate Against Human Labels	ats_evaluation.ipynb	Complete

3. Task Details, Scoring Rules, and Bugs Found

Task 4: Education Matching

- **Rule:** 50% degree level (meets required level?) + 50% field relevance (keyword overlap, scored as a percentage of the shorter field's keyword set).
- **Bug I found during testing:** My first version stripped words like 'science' and 'management' as filler, which flattened every candidate's field score to the same value. Fixed by only stripping true connector words (and, or, related, of, the).

Task 2: Skill Match Score

- **Rule:** Required and preferred skills scored separately, combined as $0.8 \times \text{required} + 0.2 \times \text{preferred}$. Required skills dominate since missing one is often a hard filter in real ATS systems.
- **Bug I found during testing:** Exact-string matching missed real matches - a resume listing 'MySQL' got no credit toward a job requiring 'SQL'. Fixed with a one-directional skill-implies map (specific tools count toward the general skill they prove).
- **Known limitation:** Doesn't split combined entries like 'Python or R' into separate options.

Task 3: Experience Matching

- **Rule:** A 3-zone curve instead of a plain ratio: full marks at or above preferred years; climbs 0.7 to 1.0 between required and preferred; drops off steeper below the required minimum.
- **Bug I found during testing:** My first version counted an ongoing class project as 6+ years of real work experience, inflating every score to a perfect 1.0. Fixed by excluding academic project entries from the professional experience total.
- **Known limitation:** Doesn't merge overlapping real job date ranges - two simultaneous roles would double-count.

Task 1: Improve Semantic Matching

- **Rule:** Resume and job description are split into 3 components (skills, experience, education), each compared separately with sentence embeddings, then combined as $0.4 \times \text{skills} + 0.4 \times \text{experience} + 0.2 \times \text{education}$ - instead of comparing the whole documents blindly like Week 2 did.
- **Known limitation:** Measures overall topical closeness, not specific requirements. In testing, a medium-match candidate scored slightly higher (0.732) than the strong-match candidate (0.713) because the wording overlapped more, even though the medium-match candidate was missing 3 required skills. This is why I don't rely on semantic score alone in Task 5.

Task 5: Build ATS Score v2

- **Rule:** $0.35 \times \text{skills} + 0.25 \times \text{experience} + 0.25 \times \text{semantic} + 0.15 \times \text{education}$. Skills weighted highest as the most concrete signal; education weighted lowest.

Task 6: Skill Gap Analysis

- **Rule:** Reuses the matched/missing lists from Task 2 and labels each missing skill as a Critical Gap (missing required) or Nice-to-Have Gap (missing preferred).
- **Bug I found during testing:** Found a formatting bug while reviewing this output: 'Agile/Scrum experience' was being squashed into 'agilescrum experience' because punctuation was deleted instead of converted to a space. Fixed, with a special case to keep 'A/B testing' readable.

4. Task 7: Evaluation Against Human Labels

I scored all three pairs myself, before looking at any AI output, then compared:

Candidate -> Job	My Score	AI Score	Gap
Al E. Gator -> Software Engineer I	95%	80.6%	-14.4
Al E. Gator (Ag. Sciences) -> Data Analyst	55%	39.3%	-15.7
Albert Gator -> Marketing Coordinator	80%	73.0%	-7.0

Overall pattern: the AI scored lower than me on all three pairs - a small gap on the medium match (-7.0), large gaps on the strong match (-14.4) and weak match (-15.7).

Software Engineer I - Me: 95% / AI: 80.6%

- I scored this candidate high partly because of the Google internship - a signal the AI has no way to measure.
- AI's skills (74%) and semantic (71%) scores were only average, pulling the total down despite perfect education and strong experience.
- Not really an AI error - employer reputation is outside what any of the four components measure.

Data Analyst - Me: 55% / AI: 39.3% (biggest gap, clearest AI weakness)

- I gave partial credit for real work experience and a relevant degree - I judged this candidate as plausibly trainable.
- AI's skill score was a flat 0% - it only recognizes exact skill-name matches, with no concept of transferable experience.
- Since skills carry the heaviest weight (35%), that single 0% dominated the final score.
- **Fix I'd make:** add partial credit for related/adjacent skills, or lower the skills weight slightly in favor of experience.

Marketing Coordinator - Me: 80% / AI: 73.0% (smallest gap)

- AI correctly caught 3 missing required skills (content creation, email marketing platform, Excel) that I may have underweighted myself.
- This looks like a fair disagreement, not an AI error - no change needed here.

5. Conclusion and Next Steps

All seven tasks are complete, tested against real resume data, and produce transparent, explainable scores instead of one opaque number. My main finding from Task 7: a single 0% skills score can dominate the final result even when a candidate is otherwise strong. Before a v2, I'd recommend:

- Add partial credit for closely related/adjacent skills instead of requiring exact name matches.
- Revisit the 35/25/25/15 weighting using a larger, more diverse set of human-labeled pairs than the three I used here.